

Reconstructable, privacy-preserving provenance for FHIR-to-large language model clinical transactions: a protocol demonstration

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Abstract

Background and Objectives: Clinical artificial intelligence (AI) systems that consume electronic health record data and emit reviewer-facing summaries rarely preserve the inputs, transformations, model configuration, evidence, safeguards, and human disposition needed to reconstruct a transaction. Storing raw clinical payloads or hidden model reasoning in an immutable log to fill that gap is unsafe. This study asks whether one Fast Healthcare Interoperability Resources (FHIR) to large language model (LLM) transaction can be reconstructed and integrity-checked without either.

Methods: Curie Audit Plane separates protected clinical content from an append-only audit chain of typed events with content digests. A completed transaction root is placed in a one-leaf Merkle envelope and signed with Ed25519. Model output is constrained to structured rationale with evidence references, uncertainty, assumptions, and missing data; hidden

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chain-of-thought fields are rejected. A versioned harness reloads persisted records, runs a separate verifier module, applies 19 labeled mutations, ablates provenance classes, probes role-based access, and measures capture overhead against an identical unrecorded clinical workflow.

Results: On synthetic FHIR Release 4 fixtures, reload-and-verify audit reconstruction completeness was 20 of 20 required fields. All 19 mutations produced a non-verified outcome (18 tampered, one incomplete), and 11 of 11 access probes matched their expected outcome. Complete capture added 103% mean latency (16.1 versus 8.6 ms; 95% interval 79–128%) and a 3.5-fold logical storage multiplier (7.2-fold allocated on disk).

Conclusions: A data/proof split makes a bounded FHIR-to-LLM handoff reconstructable without placing raw protected health information or hidden reasoning in an immutable store. The contribution integrates known integrity and interoperability primitives into application software; overhead is substantial at single-transaction scale, and clinical or population-scale evaluation remains future work.

Keywords: Fast Healthcare Interoperability Resources, Provenance, Audit trail, Large language models, Data integrity, Biomedical informatics

1. Introduction

Large language models are being attached to electronic health record workflows for summarization and review support. Healthcare organizations still need a narrower answer: given one completed transaction, can an investigator reconstruct what the model was shown, how it was configured, which evidence and safeguards were applied, and what a human reviewer decided,

7 without placing raw protected health information or hidden reasoning into
8 an immutable store?

9 This paper treats that question as application-software design for biomed-
10 ical informatics, not as generic logging and not as a clinical efficacy
11 trial. The prototype is deliberately narrow: one synthetic FHIR Re-
12 lease 4 encounter, one structured clinical-summary task, and a human AC-
13 CEPT/MODIFY/REJECT disposition. It is a protocol and software demon-
14 stration on synthetic fixtures, not a population study or a claim of readiness
15 for regulatory use.

16 The research questions, fixed before measurement, are: (i) recon-
17 structability of required provenance fields; (ii) detection of mutation, omis-
18 sion, reordering, broken references, and invalid proofs; (iii) latency and
19 storage overhead of complete capture versus an identical unrecorded clin-
20 ical workflow; (iv) console exposure of evidence, uncertainty, guardrail re-
21 sults, and disposition, demonstrated with a scripted proxy (`SCRIPTED_PROXY`)
22 rather than a blinded reviewer study; and (v) replay fidelity for a determin-
23 istic stub versus a hosted nondeterministic model.

24 The contribution is an integration: a data/proof split, typed
25 events, structured rationale without hidden chain-of-thought, FHIR
26 **Provenance/AuditEvent** projections, and reload-and-verify scoring against
27 construct-dependent analytical baselines, released as reproducible open-
28 source software. Hash chaining, Merkle trees, and Ed25519 are used as known
29 primitives, not as new cryptographic results [1, 2, 3].

30 1.1. Related work

31 Healthcare audit and provenance already have native interoperability
32 surfaces. FHIR `Provenance` and `AuditEvent` record agents, entities, and
33 access events, and W3C PROV-DM supplies a generic provenance data
34 model [4, 5, 6]. IHE Audit Trail and Node Authentication addresses node-
35 level audit trails and secure communication in clinical networks [7]. FHIR
36 implementation surveys and the FHIR security framework document how
37 these resources are used in practice [8, 9]. These artifacts are necessary in-
38 teroperability projections, but they are not, by themselves, an internal event
39 chain that can reconstruct a FHIR-to-LLM handoff: input manifests, trans-
40 formations, prompt and model identity, retrieved evidence digests, guardrail
41 outcomes, and human disposition.

42 Tamper-evident logging has a long cryptographic literature. Haber and
43 Stornetta time-stamped documents with hash-linked records; Merkle trees
44 supply inclusion proofs; Schneier and Kelsey and Crosby and Wallach devel-
45 oped hash-chained and efficient tamper-evident logs; and Certificate Trans-
46 parency uses Merkle trees for publicly auditable logs [1, 2, 10, 11, 12]. This
47 prototype composes those primitives in a local append-only store. It is not
48 a distributed ledger and is not evaluated against blockchain-in-healthcare
49 architectures [13].

50 Clinical LLM evaluation has focused mainly on accuracy, fairness, and the
51 limits of electronic-health-record foundation models [14, 15]. Adapted mod-
52 els can outperform experts on clinical text summarization, and LLMs encode
53 substantial clinical knowledge, but those results do not address transaction
54 auditability [16, 17]. A systematic review of 519 health-care LLM studies

55 found that almost all used accuracy as the primary dimension and that de-
56 ployment considerations were rarely measured [15]. Clinical AI governance
57 work, including model cards, model-facts labels, and good machine learn-
58 ing practice, describes intended use and performance [18, 19, 20]; it does
59 not record the inputs, evidence, guardrails, and human disposition of one
60 completed transaction.

61 **2. Methods**

62 *2.1. System boundary*

63 Curie Audit Plane observes a synthetic FHIR-to-LLM workflow (Fig. 1).
64 Protected payloads (assembled context, model request and response,
65 reviewer-modified text, and free-form comments) are stored in an authorized
66 content store. The immutable audit store holds metadata, opaque references,
67 schemas, and digests. Reviewer comments on the chain are limited to a con-
68 trolled category; detailed text is protected. Provider exception text is neither
69 stored nor fingerprinted. Patient and resource identifiers in immutable meta-
70 data are opaque tokens; a protected identity map is required for authorized
71 reconstruction.

72 *2.2. Event model and integrity*

73 A successful transaction emits typed events from input manifestation
74 through human action and an integrity-proof commitment. Events are canon-
75 ically hashed and chained. The completed transaction root is represented as
76 a one-leaf Merkle envelope and signed with Ed25519; the measured pipeline
77 does not aggregate multiple transaction roots into one batch. Verification

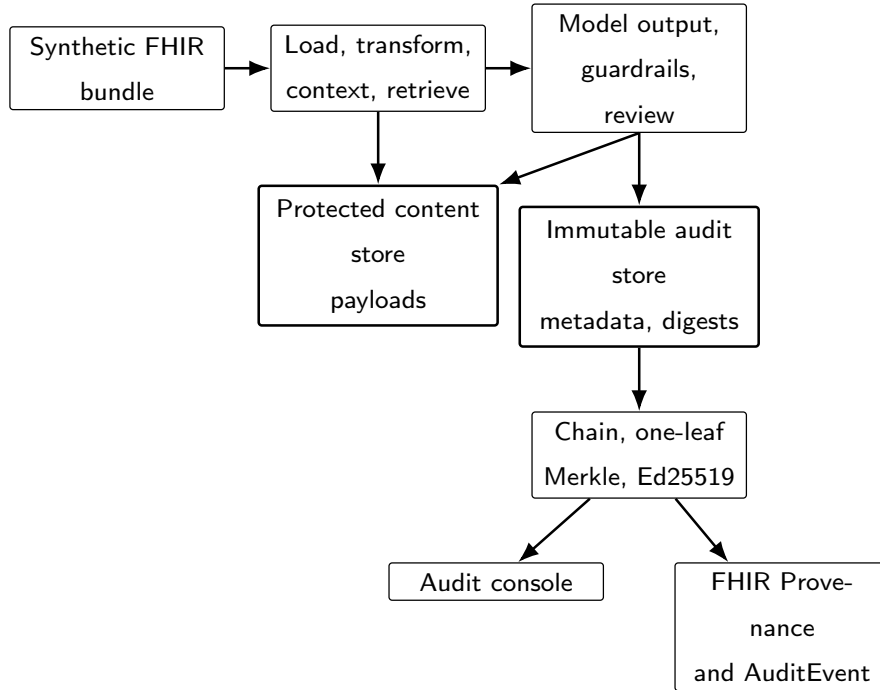


Figure 1: System boundary. Protected clinical payloads stay outside the immutable audit store. The measured path signs one completed transaction root per Merkle envelope.

78 reports **VERIFIED**, **INCOMPLETE**, **TAMPERED**, or **FAILED**. The verifier is a separate
 79 module in the same repository, run after persisted SQLite and protected-
 80 content records are reloaded; it is not an external third-party audit. Access
 81 and export events are written to a separate hash-chained access-audit stream,
 82 so reading a sealed transaction does not alter its clinical proof. FHIR Re-
 83 lease 4 **Provenance** and **AuditEvent** resources are documented projections
 84 of the internal model, not a validated Implementation Guide profile.

85 2.3. Structured output and guardrails

86 Model output must validate as structured rationale. Guardrails record
 87 schema validity, evidence-reference coverage, uncertainty presence, a syn-

88 thetic identifier scan, and policy action. Blocking results require an explicit
89 override policy before ACCEPT. Hidden chain-of-thought fields are rejected
90 at the schema boundary.

91 *2.4. Software implementation*

92 The system is implemented in Python with a FastAPI service, SQLite au-
93 dit and protected-content stores, and a React audit console that presents an
94 event timeline, formatted and raw JSON, verification scope, and an artifact-
95 count flow view. API endpoints are role-gated (reviewer, investigator, ad-
96 min) with locally generated bearer tokens. The reproducible entry point is
97 `curie-audit-plane evaluate`.

98 *2.5. Evaluation protocol*

99 Reports use schema `curie-evaluation.v1.1` and record experiment
100 metadata (git commit, dirty-tree flag, fixture alias and digest, provider,
101 model, prompt version, decoding parameters, endpoint class, seed, and com-
102 mand template) without absolute local paths.

103 The headline metric, audit reconstruction completeness (ARC), is the
104 fraction of 20 required provenance fields present after reloading per-
105 sisted records, counted only when the verifier succeeds (report field
106 `independently_verified_arc`). Field-presence completeness on the in-
107 memory object is reported separately. Mutation detection uses 19 labeled
108 cases; a case is detected when the verifier returns any non-VERIFIED sta-
109 tus, and the specific status is reported. Replay is classified as EXACT_MATCH,
110 EQUIVALENT, or DIVERGENT.

111 Overhead compares the complete plane with an identical unrecorded clin-
 112 ical workflow (load, transform, context, retrieve, complete, guardrails, and an
 113 in-memory ACCEPT disposition written only to JSONL) on isolated stores,
 114 with one warmup and 30 timed repeats. The unrecorded path emits no audit
 115 events, Merkle proofs, or signatures.

$$\text{overhead}_{\text{latency}} = (T_{\text{plane}} - T_{\text{no_audit}})/T_{\text{no_audit}}, \quad (1)$$

$$\text{storage multiplier} = B_{\text{plane}}/B_{\text{no_audit}}. \quad (2)$$

116 Allocated bytes are on-disk file sizes, including SQLite page allocation. Log-
 117 ical bytes are UTF-8 octet lengths of serialized event JSON, transaction-row
 118 payloads, and protected-content files, excluding page padding. Rate metrics
 119 use Wilson intervals.

120 Ablations omit input manifests, transformations, model and prompt
 121 metadata, evidence references, cryptographic proofs, or human provenance.
 122 Access-control probes record allowed and denied HTTP outcomes for re-
 123 viewer, investigator, admin, output, content, export, missing transaction,
 124 and global list scope. The comparison recorders (application JSONL, hash-
 125 only JSONL, and a FHIR R4 projection) are analytical emulations built for
 126 this study and scored against the same required-field inventory; they are not
 127 independently shipped products.

128 The default 50-encounter cohort rewrites one synthetic fixture; it tests
 129 repeatability of a single workflow, not a population. Optional Synthea
 130 slices [21] are excluded from measured evidence because generator version,
 131 seed, module set, and command line are not pinned.

132 2.6. Threat model

133 Adversaries may mutate, delete, or reorder events; substitute proofs or
134 keys; present unsupported findings; or attempt to place protected health
135 information in reviewer comments, provider errors, research exports, or end-
136 point configuration. The prototype assumes synthetic data, local keys, and
137 an approved loopback model endpoint. It does not protect against compro-
138 mised signing keys or a malicious protected-content store operator.

139 3. Results

140 All numbers are from the deterministic-stub evaluation report in
141 `evaluation-results/` (schema `curie-evaluation.v1.1`, git commit
142 `0dcf801`, clean working tree), archived under tag `jbhi-eval-20260828`. At
143 that commit the repository had 146 Python tests, 94.59% statement cover-
144 age, and a clean linter pass; these are software quality indicators, not clinical
145 performance.

146 3.1. Reconstructability and tamper detection

147 ARC was 20/20 required fields on the clean fixture transaction after the
148 verifier succeeded; field-presence completeness on the in-memory object was
149 also 20/20, and required-event completeness was 11/11. All 19 labeled mu-
150 tations produced a non-`VERIFIED` result: 18 were classified `TAMPERED`, and
151 one substituted content reference was classified `INCOMPLETE` because the ref-
152 erenced protected content was absent. The false-tamper rate over three clean
153 runs was 0. Verification of the clean transaction took 1.55 ms. Across the
154 50-encounter cohort, ARC, required-event completeness, and human-action

Table 1: Comparison recorders on the synthetic fixture. Scores use this paper’s required-field inventory and are construct-dependent.

Recorder	ARC	Tamper detection	Queryability
Application JSONL	0.10	0.00	0.00
Hash-only JSONL	0.15	1.00	0.00
FHIR R4 projection	0.20	0.00	0.25
Complete plane	1.00	1.00	1.00

capture were each 50/50 (Wilson 95% interval 0.93–1.00), with mean verification latency 1.21 ms.

The complete plane reconstructs more required fields and detects the standard model-event mutation that application logs and FHIR projections do not (Table 1). Hash-only logging detects digest mismatch without chain or signature guarantees. These gaps are expected by construction: the comparison recorders were not designed to capture the same field set.

3.2. Ablations

Table 2 reports ARC after omitting one provenance class. Omitting cryptographic proofs has the largest effect (0.75), followed by human provenance (0.85) and model and prompt metadata (0.90). These are field-presence ablations, not clinical-efficacy ablations.

3.3. Access control

Eleven role probes returned the predeclared HTTP status (11/11; Wilson 95% interval 0.74–1.00): reviewer read and output allowed; reviewer export and investigator output or content denied; investigator verify and export

Table 2: Audit reconstruction completeness after omitting a provenance class.

Condition	ARC	Δ from full
Full recorded provenance	1.00	0.00
Omit input manifests	0.95	0.05
Omit transformations	0.95	0.05
Omit model and prompt metadata	0.90	0.10
Omit evidence references	0.95	0.05
Omit cryptographic proofs	0.75	0.25
Omit human provenance	0.85	0.15

Table 3: Capture overhead versus an identical unrecorded workflow ($n = 30$ paired runs for latency).

Measure	Plane	Unrecorded	Ratio
Mean latency (ms)	16.13	8.59	+103.3%
Allocated storage (bytes)	81 885	11 373	7.20×
Logical storage (bytes)	39 454	11 373	3.47×

171 allowed; admin content allowed; unauthenticated list denied (401); missing
 172 transaction reported as missing (404); and global list allowed.

173 3.4. Overhead and replay

174 Complete capture roughly doubled latency (Table 3; mean paired rela-
 175 tive overhead +103.3%, 95% interval +78.8% to +127.8%). The allocated
 176 audit file was a single 65 536-byte SQLite page holding 23 105 logical bytes
 177 of events and transaction rows, so page allocation inflates the on-disk mul-

178 tiplier; the logical multiplier (3.47) is the less confounded figure. These
179 results do not support a claim of low overhead at single-transaction scale.
180 Deterministic-stub same-prompt replay was exact, and prompt-substitution
181 replay was DIVERGENT by construction. The one hosted same-prompt replay
182 observation was DIVERGENT ($n = 1$); no hosted replay rate is estimated.

183 A 16-arm scenario matrix exercised ACCEPT/MODIFY/REJECT,
184 forced and natural WARN/BLOCK guardrail outcomes, provider failure, a
185 sparse encounter, MODIFY evidence, prompt-substitution replay, and access
186 audit. It demonstrates workflow coverage rather than sampling a population;
187 the two unpinned Synthea arms are excluded from the measured tables. Fig. 2
188 shows the console surfaces used in the scripted reconstruction demonstration.

189 4. Discussion

190 In an electronic health record setting the plane would sit beside an existing
191 FHIR access path rather than replace it. A hospital integration would con-
192 tinue to enforce resource authorization in the clinical system, store protected
193 payloads under that authority, and emit the typed audit events described
194 here, with FHIR **Provenance** and **AuditEvent** as the interoperability pro-
195 jection for downstream governance tools. The design keeps the LLM off any
196 deterministic alert-firing path, and reconstructability does not imply clinical
197 correctness.

198 The ablations indicate where the reconstruction value lies: cryptographic
199 proofs and human provenance account for the largest share of required fields,
200 which is the information ordinary application logs and FHIR projections
201 most often lack. The overhead results are the main practical cost. At one

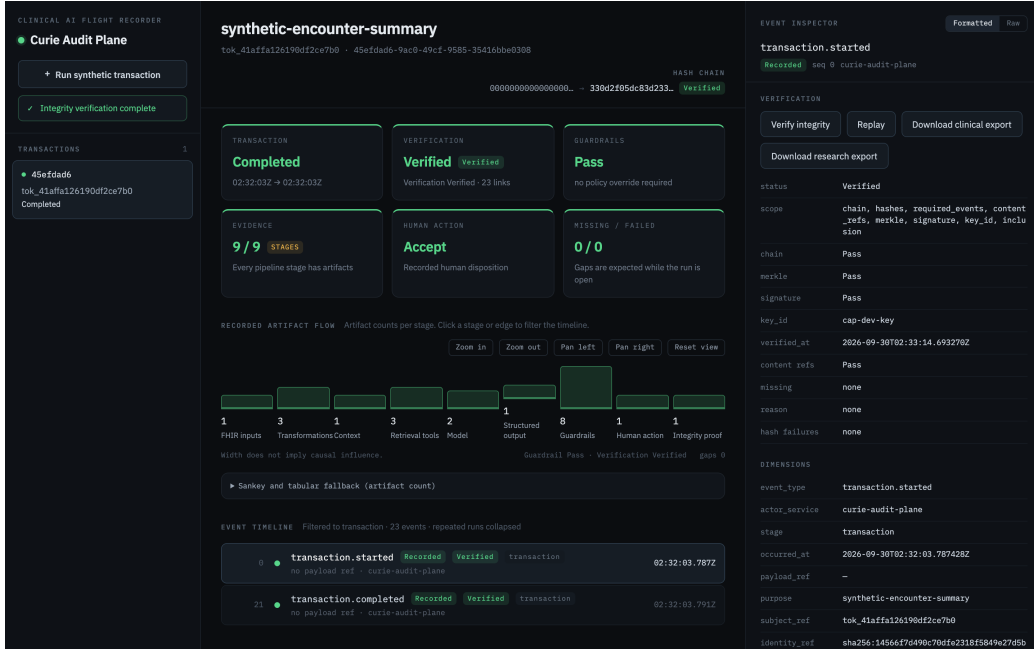


Figure 2: Audit console on a synthetic transaction after integrity verification: overview tiles, recorded artifact-count flow (width does not imply causal influence), event timeline, and verification scope with chain, Merkle, signature, and content-reference checks. The subject is shown only as an opaque token.

transaction per Merkle envelope, fixed per-transaction work (signing, chain-
ing, page allocation) dominates; batching multiple roots per envelope is the
obvious amortization path but was not evaluated.

4.1. Limitations

This is a single-workflow protocol demonstration on synthetic FHIR fix-
tures. The cohort rewrites one fixture and is not 50 independent patients.
Baseline scores are construct-dependent, measured against analytical emu-
lations rather than shipped logging products. The verifier is in the same
repository, not an external audit. Each Merkle envelope holds one transac-

tion, so batching and multi-transaction storage amortization are unreported. Latency with $n = 30$ is an overhead estimate, not a capacity or tail-latency study. FHIR projections are mapping rules without Implementation Guide validation. Console reconstruction was demonstrated with a scripted proxy, not with clinicians or auditors.

4.2. Conclusions

A privacy-preserving provenance plane can reconstruct a bounded synthetic FHIR-to-LLM transaction from persisted records, detect a defined mutation suite, and keep hidden reasoning and free-form reviewer text out of the immutable chain, at the cost of roughly doubled latency and a 3.5-fold logical storage multiplier at single-transaction scale. The result integrates known primitives under a clinical data/proof split; it is not a new integrity scheme or a population evaluation. Next steps are independent synthetic patients at scale, batched envelopes, hosted-model replay characterization, and a reconstruction study with clinical auditors.

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Ethics

No human subjects or identifiable clinical records were used. The evaluation uses synthetic FHIR templates; therefore no institutional review board approval was required.

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236 **Declaration of competing interest**

237 The author declares no known competing financial interests or personal
238 relationships that could have appeared to influence the work reported in this
239 paper.

240 **Data and code availability**

241 Source code, synthetic fixtures, and versioned evaluation artifacts are
242 openly available at <https://github.com/sreeram843/curie-audit-plane>
243 under tag `jbhi-eval-20260828`. No patient data were used.

244 **CRedit authorship contribution statement**

245 **Satya Venkata Ranga Janaki Sriram Mentey:** Conceptualization,
246 Methodology, Software, Validation, Formal analysis, Investigation, Data cu-
247 ration, Writing – original draft, Writing – review & editing, Visualization.

248 **Declaration of generative AI and AI-assisted technologies in the** 249 **writing process**

250 During the preparation of this work the author used AI coding agents
251 to draft portions of the software and documentation and to assist with
252 manuscript editing. All evaluation metrics were produced by the versioned

253 Python harness. After using these tools, the author reviewed and edited
254 the content as needed and takes full responsibility for the content of the
255 publication.

256 References

- 257 [1] S. Haber, W. S. Stornetta, How to time-stamp a digital document, Jour-
258 nal of Cryptology 3 (2) (1991) 99–111. doi:10.1007/BF00196791.
- 259 [2] R. C. Merkle, A digital signature based on a conventional encryption
260 function, in: Advances in Cryptology—CRYPTO '87, Springer, 1988,
261 pp. 369–378. doi:10.1007/3-540-48184-2_32.
- 262 [3] S. Josefsson, I. Liusvaara, Edwards-Curve Digital Signature Algo-
263 rithm (EdDSA), IETF RFC 8032, [https://www.rfc-editor.org/rfc/](https://www.rfc-editor.org/rfc/rfc8032)
264 [rfc8032](https://www.rfc-editor.org/rfc/rfc8032) (Jan. 2017). doi:10.17487/RFC8032.
- 265 [4] HL7 International, Provenance, FHIR Release 4, [http://hl7.org/](http://hl7.org/fhir/R4/provenance.html)
266 [fhir/R4/provenance.html](http://hl7.org/fhir/R4/provenance.html) (2019).
- 267 [5] HL7 International, Auditevent, FHIR Release 4, [http://hl7.org/](http://hl7.org/fhir/R4/auditevent.html)
268 [fhir/R4/auditevent.html](http://hl7.org/fhir/R4/auditevent.html) (2019).
- 269 [6] World Wide Web Consortium, PROV-DM: The PROV data model,
270 Tech. rep., World Wide Web Consortium, W3C Recommendation.
271 <https://www.w3.org/TR/prov-dm/> (Apr. 2013).
- 272 [7] Integrating the Healthcare Enterprise, IHE IT infrastructure technical
273 framework volume 1: Integration profiles, audit Trail and Node Au-

- 274 thentication (ATNA). [https://profiles.ihe.net/ITI/TF/Volume1/](https://profiles.ihe.net/ITI/TF/Volume1/ch-9.html)
275 [ch-9.html](https://profiles.ihe.net/ITI/TF/Volume1/ch-9.html) (2024).
- 276 [8] M. Ayaz, M. F. Pasha, M. Y. Alzahrani, R. Budiarto, D. Stiawan,
277 The fast health interoperability resources (FHIR) standard: System-
278 atic literature review of implementations, applications, challenges and
279 opportunities, JMIR Medical Informatics 9 (7) (2021) e21929. doi:
280 [10.2196/21929](https://doi.org/10.2196/21929).
- 281 [9] HL7 International, FHIR security, FHIR Release 4, [http://hl7.org/](http://hl7.org/fhir/R4/security.html)
282 [fhir/R4/security.html](http://hl7.org/fhir/R4/security.html) (2019).
- 283 [10] B. Schneier, J. Kelsey, Cryptographic support for secure logs on un-
284 trusted machines, in: Proceedings of the 7th USENIX Security Sympos-
285 sium, San Antonio, TX, USA, 1998, pp. 53–62.
- 286 [11] S. A. Crosby, D. S. Wallach, Efficient data structures for tamper-evident
287 logging, in: Proceedings of the 18th USENIX Security Symposium, Mon-
288 treal, QC, Canada, 2009, pp. 317–334.
- 289 [12] B. Laurie, A. Langley, E. Kasper, Certificate transparency, IETF RFC
290 6962, <https://www.rfc-editor.org/rfc/rfc6962> (Jun. 2013). doi:
291 [10.17487/RFC6962](https://doi.org/10.17487/RFC6962).
- 292 [13] C. C. Agbo, Q. H. Mahmoud, J. M. Eklund, Blockchain technology
293 in healthcare: A systematic review, Healthcare 7 (2) (2019) 56. doi:
294 [10.3390/healthcare7020056](https://doi.org/10.3390/healthcare7020056).
- 295 [14] M. Wornow, Y. Xu, R. Thapa, B. Patel, E. Steinberg, S. Fleming, M. A.
296 Pfeffer, J. Fries, N. H. Shah, The shaky foundations of large language

- models and foundation models for electronic health records, *npj Digital Medicine* 6 (1) (2023) 135. [doi:10.1038/s41746-023-00879-8](https://doi.org/10.1038/s41746-023-00879-8).
- [15] S. Bedi, Y. Liu, L. Orr-Ewing, D. Dash, S. Koyejo, A. Callahan, J. A. Fries, M. Wornow, A. Swaminathan, L. S. Lehmann, H. J. Hong, M. Kashyap, A. R. Chaurasia, N. R. Shah, K. Singh, T. Tazbaz, A. Milstein, M. A. Pfeffer, N. H. Shah, Testing and evaluation of health care applications of large language models: A systematic review, *JAMA* 333 (4) (2025) 319–328. [doi:10.1001/jama.2024.21700](https://doi.org/10.1001/jama.2024.21700).
- [16] D. Van Veen, C. Van Uden, L. Blankemeier, J.-B. Delbrouck, A. Aali, C. Bluethgen, A. Pareek, M. Polacin, E. P. Reis, A. Seehofnerová, N. Rohatgi, P. Hosamani, W. Johnson, C. Langlotz, A. S. Chaudhari, Adapted large language models can outperform medical experts in clinical text summarization, *Nature Medicine* 30 (2024) 1134–1142. [doi:10.1038/s41591-024-02855-5](https://doi.org/10.1038/s41591-024-02855-5).
- [17] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, Large language models encode clinical knowledge, *Nature* 620 (2023) 172–180. [doi:10.1038/s41586-023-06291-2](https://doi.org/10.1038/s41586-023-06291-2).
- [18] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, T. Gebru, Model cards for model reporting, in: *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2019, pp. 220–229. [doi:10.1145/3287560.3287596](https://doi.org/10.1145/3287560.3287596).
- [19] M. P. Sendak, M. Gao, N. Brajer, S. Balu, Presenting machine learning

- 319 model information to clinical end users with model facts labels, npj
320 Digital Medicine 3 (2020) 41. doi:10.1038/s41746-020-0253-3.
- 321 [20] U.S. Food and Drug Administration, Health Canada, Medicines and
322 Healthcare products Regulatory Agency, Good machine learning prac-
323 tice for medical device development: Guiding principles, <https://www.fda.gov/media/153241/download> (Oct. 2021).
- 324
- 325 [21] J. Walonoski, M. Kramer, J. Nichols, A. Quina, C. Moesel, D. Hall,
326 C. Duffett, K. Dube, T. Gallagher, S. McLachlan, Synthea: An ap-
327 proach, method, and software mechanism for generating synthetic pa-
328 tients and the synthetic electronic health care record, Journal of the
329 American Medical Informatics Association 25 (3) (2018) 230–238. doi:
330 10.1093/jamia/ocx079.